

Multi-Mode Microphone with I²S Digital Output

GENERAL DESCRIPTION

The ICS-43434 is digital I²S output bottom port microphone. The complete ICS-43434 solution consists of a MEMS sensor, signal conditioning, an analog-to-digital converter, decimation and antialiasing filters, power management, and an industry standard 24-bit I²S interface. The I²S interface allows the ICS-43434 to connect directly to digital processors, such as DSPs and microcontrollers, without the need for an audio codec in the system.

The ICS-43434 has multiple modes of operation: High Performance, Low Power (AlwaysOn) and Sleep. The ICS-43434 has high SNR and 120 dB SPL AOP in all operational modes.

The ICS-43434 has a high SNR of 65 dBA and a wideband frequency response. The sensitivity tolerance of the ICS-43434 is ± 1 dB, which enables high-performance microphone arrays without the need for system calibration.

The ICS-43434 is available in a small 3.50 mm $\times$ 2.65 mm $\times$ 0.98 mm surface-mount package. The ICS-43434 is function-compatible with the ICS-43432 while providing equivalent electro-acoustic performance at lower power consumption and in a smaller package.

APPLICATIONS

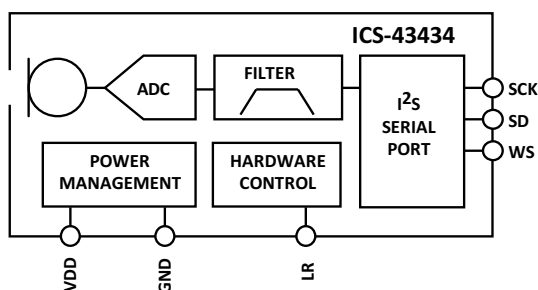
- Wearables
- Smart Televisions
- Remote Controls
- IoT Devices
- Teleconferencing Systems
- Gaming Consoles
- Security Systems

FEATURES

SPEC	HIGH PERFORMANCE MODE	LOW-POWER MODE
Sensitivity	-26 dB FS ± 1 dB	-26 dB FS ± 1 dB
SNR	65 dBA	64 dBA
Current	490 μ A	230 μ A
AOP	120 dB SPL	120 dB SPL
Sample Rate	23 – 51.6 kHz	6.25 – 18.75 kHz

- Digital I²S interface with high precision 24-bit data
- Wide frequency response from 60 Hz to 20 kHz
- High power supply rejection: -100 dB FS
- Small 3.50 mm $\times$ 2.65 mm $\times$ 0.98 mm surface-mount package
- Compatible with Sn/Pb and Pb-free solder processes
- RoHS/WEEE compliant

FUNCTIONAL BLOCK DIAGRAM



ORDERING INFORMATION

PART	TEMP RANGE	PACKAGING
ICS-43434	-40°C to +85°C	13" Tape & Reel
EV_IC3-43434-FX	—	

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SPECIFICATIONS

TABLE 1. ACOUSTICAL/ELECTRICAL CHARACTERISTICS – GENERAL

$T_A = 25^\circ\text{C}$, $V_{DD} = 1.8$ to 3.3 V, $f_s = 48$ kHz, $C_{LOAD} = 30$ pF unless otherwise noted. Typical specifications are not guaranteed.

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
PERFORMANCE						
Directionality		Omni				
Output Polarity	Input acoustic pressure vs. output data	Non-Inverted				
Supply Voltage (V_{DD})		1.65		3.63	V	
Sleep Mode Current (I_S)	$f_s < 3.125$ kHz		12	20	μA	

TABLE 2. ACOUSTICAL/ELECTRICAL CHARACTERISTICS – HIGH-PERFORMANCE MODE

$T_A = 25^\circ\text{C}$, $V_{DD} = 1.8$ to 3.3 V, $f_s = 48$ kHz, $C_{LOAD} = 30$ pF unless otherwise noted. Typical specifications are not guaranteed.

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
Sensitivity	1 kHz, 94 dB SPL	-27	-26	-25	dB FS	1
Signal-to-Noise Ratio (SNR)	20 kHz bandwidth, A-weighted		65		dBA	
Equivalent Input Noise (EIN)	20 kHz bandwidth, A-weighted		29		dBA SPL	
Dynamic Range	Derived from EIN and acoustic overload point		91		dB	
Total Harmonic Distortion (THD)	105 dB SPL		0.2	1	%	
Power Supply Rejection (PSR)	217 Hz, 100 mV p-p square wave superimposed on $V_{DD} = 1.8$ V, A-weighted		-99		dB FS	
Power Supply Rejection—Swept Sine	1 kHz sine wave, $V_{DD} = 1.8$ V		-106		dB FS	
Acoustic Overload Point	10% THD		120		dB SPL	
Noise Floor	20 Hz to 20 kHz, A-weighted, rms		-90		dB FS	
Supply Current (I_S)	$V_{DD} = 1.8$ V, no load		490	550	μA	

Note 1: Sensitivity is relative to the RMS level of a sine wave with positive amplitude equal to 100% 1s density and negative amplitude equal to 0% 1s density.

TABLE 3. ACOUSTICAL/ELECTRICAL CHARACTERISTICS – LOW-POWER MODE

$T_A = 25^\circ\text{C}$, $V_{DD} = 1.8$ to 3.3 V, $f_s = 16$ kHz, $C_{LOAD} = 30$ pF unless otherwise noted. Typical specifications are not guaranteed.

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
Sensitivity	1 kHz, 94 dB SPL	–27	–26	–25	dB FS	1
Signal-to-Noise Ratio (SNR)	20 kHz bandwidth, A-weighted		64		dBA	
Equivalent Input Noise (EIN)	20 kHz bandwidth, A-weighted		30		dBA SPL	
Dynamic Range	Derived from EIN and acoustic overload point		90		dB	
Total Harmonic Distortion (THD)	105 dB SPL		0.2	1	%	
Power Supply Rejection (PSR)	217 Hz, 100 mV p-p square wave superimposed on $V_{DD} = 1.8$ V, A-weighted		–98		dB FS	
Power Supply Rejection—Swept Sine	1 kHz sine wave, $V_{DD} = 1.8$ V		–100		dB FS	
Acoustic Overload Point	10% THD		120		dB SPL	
Noise Floor	20 kHz bandwidth, A-weighted, rms		–90		dB FS	
Supply Current (I_S)	$V_{DD} = 1.8$ V, no load		230	300	μA	

Note 1: Sensitivity is relative to the RMS level of a sine wave with positive amplitude equal to 100% 1s density and negative amplitude equal to 0% 1s density.

TABLE 4. DIGITAL FILTER CHARACTERISTICS

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
Group Delay	Acoustic input to digital output – includes filter and I ² S serial output		$2/f_s$		sec.	
Pass Band Ripple				± 0.3	dB	
Stop Band Attenuation			58		dB	
Pass Band	$f_s = 48$ kHz		20		kHz	

TABLE 5. I²S DIGITAL INPUT/OUTPUT–40°C < T_A < +85°C, 1.8 V < V_{DD} < 3.3 V, unless otherwise noted.

PARAMETER	CONDITIONS	MIN	MAX	UNITS	NOTES
MODE SWITCHING					
Sleep Time	Time from f _S falling < 3.125 kHz		1	ms	
Wake-Up Time	High-Performance mode, Sleep Mode to f _{WS} > 21.875 kHz, output within 1 dB of final sensitivity, power on		20	ms	
Wake-Up Time	Low-Power Mode, Sleep Mode to f _{WS} > 6.25 kHz, output within 1 dB of final sensitivity, power on		20	ms	
Switching time	Between Low-Power and High-Performance Modes		10	ms	
INPUT/OUTPUT					
SCK period (t _{SCP})	Input clock period	303	2500	ns	
Sampling Frequency (f _S)	Sleep Mode		3.125	kHz	
	Low-Power Mode	6.25	18.75	kHz	
	High-Performance Mode	23	51.6	kHz	
SCK high (t _{SCH})		50		ns	
SCK low (t _{SCL})		50		ns	
WS setup (t _{WSS})		0		ns	
WS hold (t _{WSH})		20		ns	
SCK Duty Cycle		40	60	%	
SD Data Valid (t _{SDV})	From SCK falling to valid SD data		75	ns	
SD Data Disable (t _{SDD})	From SCK falling to SD output tristated		76	ns	
t _{RISE}	SCK rise time (10% to 90% level)		25	ns	1
t _{FALL}	SCK fall time (90% to 10% level)		25	ns	1
DIGITAL INPUT					
Voltage Input Low (V _{IL})		0	0.3 × V _{DD}	V	
Voltage Input High (V _{IH})		0.7 × V _{DD}	V _{DD}	V	
SD DIGITAL OUTPUT					
Voltage Output Low (V _{OL})		0	0.35 × V _{DD}	V	
Voltage Output High (V _{OH})		0.65 × V _{DD}	V _{DD}	V	
Maximum Load	f _S = 48 kHz		85	pF	

Note 1: Guaranteed by design

TIMING DIAGRAM

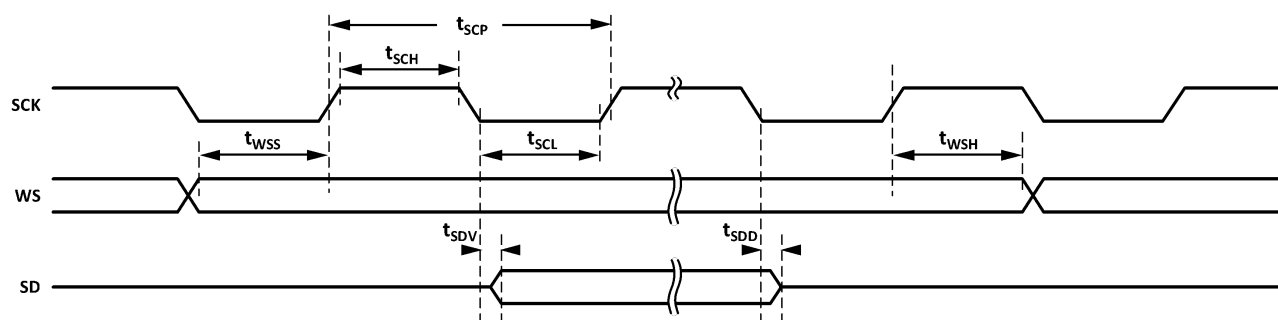


Figure 1. Serial Data Port Timing

ABSOLUTE MAXIMUM RATINGS

Stress above those listed as Absolute Maximum Ratings may cause permanent damage to the device. These are stress ratings only and functional operation of the device at these conditions is not implied. Exposure to the absolute maximum ratings conditions for extended periods may affect device reliability.

TABLE 6. ABSOLUTE MAXIMUM RATINGS

PARAMETER	RATING
Supply Voltage (V_{DD})	-0.3 V to +3.63 V
Digital Pin Input Voltage	-0.3 V to $V_{DD} + 0.3$ V or 3.63 V, whichever is less
Sound Pressure Level	160 dB
Mechanical Shock	10,000 g
Vibration	Per MIL-STD-883 Method 2007, Test Condition B
Temperature Range	
Biased	-40°C to +85°C
Storage	-55°C to +150°C

ESD CAUTION



ESD (electrostatic discharge) sensitive device. Charged devices and circuit boards can discharge without detection. Although this product features patented or proprietary protection circuitry, damage may occur on devices subjected to high energy ESD. Therefore proper ESD precautions should be taken to avoid performance degradation or loss of functionality.

SOLDERING PROFILE

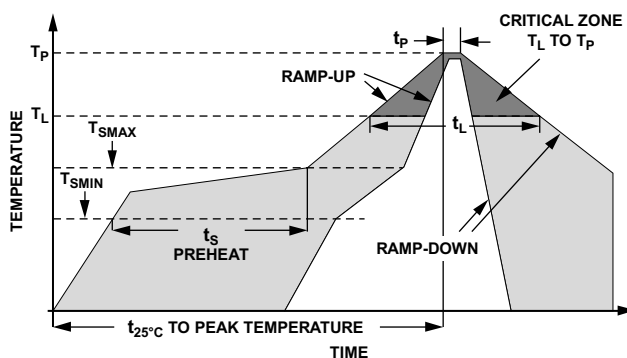


Figure 2. Recommended Soldering Profile Limits

TABLE 7. RECOMMENDED SOLDERING PROFILE

PROFILE FEATURE		Sn63/Pb37	Pb-Free
Average Ramp Rate (T_L to T_P)		1.25°C/sec max	1.25°C/sec max
Preheat	Minimum Temperature (T_{SMIN})	100°C	100°C
	Minimum Temperature (T_{SMIN})	150°C	200°C
	Time (T_{SMIN} to T_{SMAX}), t_s	60 sec to 75 sec	60 sec to 75 sec
Ramp-Up Rate (T_{SMAX} to T_L)		1.25°C/sec	1.25°C/sec
Time Maintained Above Liquidous (t_L)		45 sec to 75 sec	~50 sec
Liquidous Temperature (T_L)		183°C	217°C
Peak Temperature (T_P)		215°C +3°C/-3°C	260°C +0°C/-5°C
Time Within +5°C of Actual Peak Temperature (t_p)		20 sec to 30 sec	20 sec to 30 sec
Ramp-Down Rate		3°C/sec max	3°C/sec max
Time +25°C ($t_{25^\circ\text{C}}$) to Peak Temperature		5 min max	5 min max

*The reflow profile in Table 7 is recommended for board manufacturing with InvenSense MEMS microphones. All microphones are also compatible with the J-STD-020 profile

PIN CONFIGURATIONS AND FUNCTION DESCRIPTIONS

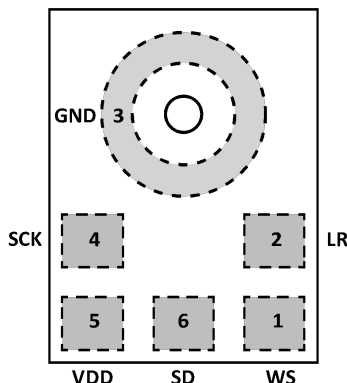


Figure 3. Pin Configuration (top view, terminal side down)

TABLE 8. PIN FUNCTION DESCRIPTIONS

PIN	NAME	TYPE	FUNCTION
1	WS	Input	Serial Data-Word Select for I ² S Interface.
2	LR	Input	Left/Right channel select. When set low, the microphone outputs its signal in the left channel of the I ² S frame. When set high, the microphone outputs its signal in the right channel.
3	GND	Ground	Ground. Connect to ground on the PCB.
4	SCK	Input	Serial Data Clock for I ² S Interface.
5	VDD	Power	Power, 1.65 to 3.63 V. This pin should be decoupled to GND with a 0.1 μ F capacitor.
6	SD	Output	Serial Data Output for I ² S Interface. This pin tristates when not actively driving the appropriate output channel. The SD trace should have a 100 k Ω pull-down resistor to discharge the line during the time that all microphones on the bus have tristated their outputs.

TYPICAL PERFORMANCE CHARACTERISTICS

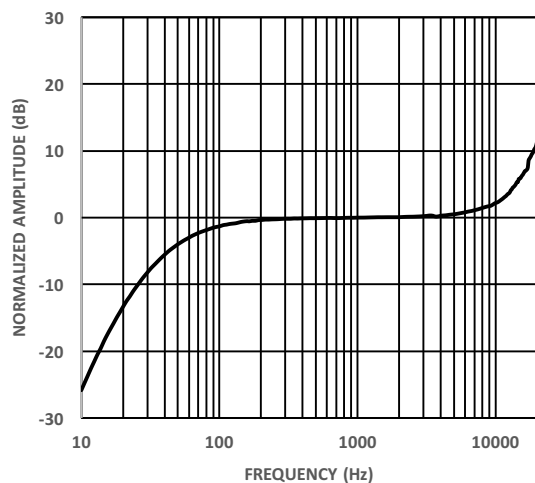


Figure 4. Typical Frequency Response (Measured)

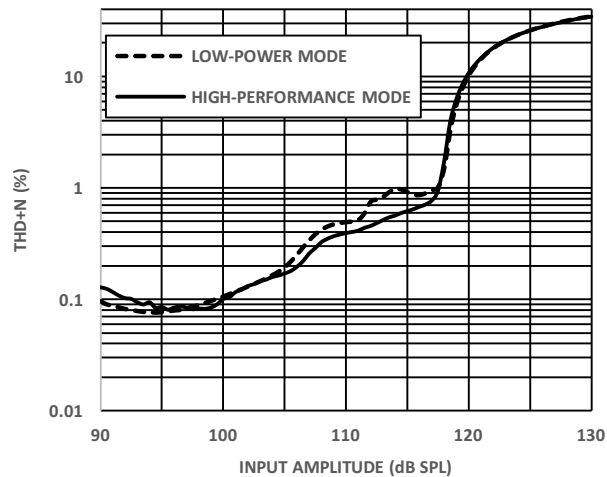


Figure 5. Total Harmonic Distortion + Noise (THD+N) vs. Input SPL

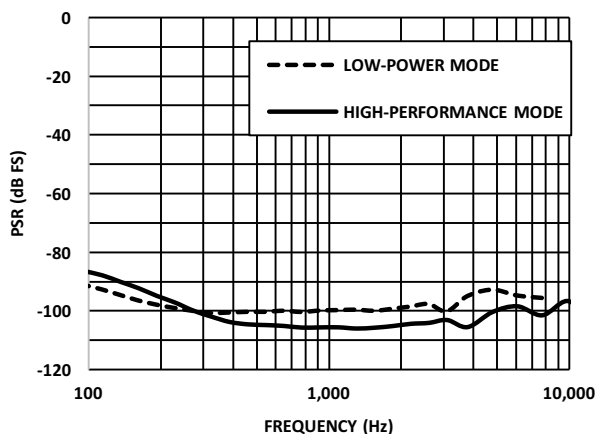


Figure 6. PSR vs. Frequency, 100 mV p-p Swept Sine Wave

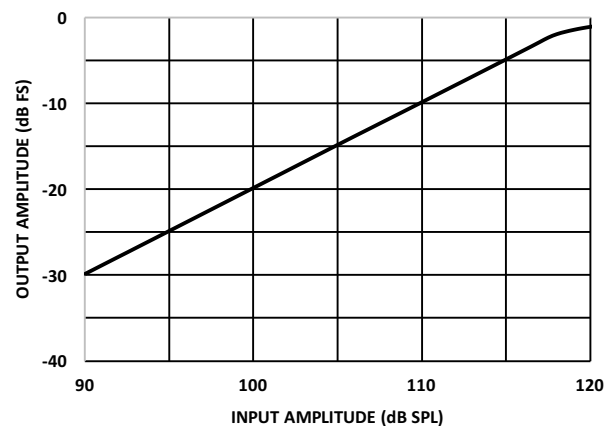


Figure 7. Linearity

THEORY OF OPERATION

POWER MANAGEMENT

The ICS-43434 has three power states: high-performance mode, low-power mode and standby mode.

Startup and Normal Operation

The start-up time of the ICS-43434 is less than 20 ms. The I²S data from the microphone is valid to be used as soon as the data is being output. The part is in normal operation (high-performance and low-power modes) when SCK and WS are active.

Standby Mode

The microphone enters standby mode when the frequency of SCK falls below about 200 kHz. It is recommended to enter standby mode by stopping both the SCK and WS clock signals and pulling those signals to ground to avoid drawing current through the WS pin's internal pull-down resistor. The timing for exiting standby mode is the same as normal startup.

It is not recommended to supply active clocks (WS and SCK) to the ICS-43434 while there is no power supplied to VDD. Doing this continuously turns on ESD protection diodes, which may affect long-term reliability of the microphone.

SYNCHRONIZING MICROPHONES

Stereo ICS-43434 microphones are synchronized by the WS signal, so audio captured from two microphones sharing the same clock will be in sync. The two microphones will synchronously sample the acoustic signals at the beginning of the I²S frame (WS falling edge).

I²S DATA INTERFACE

The slave serial data port's format is I²S, 24-bit, twos complement. There must be 64 SCK cycles in each WS stereo frame. The LR control pin determines whether the ICS-43434 outputs data in the left or right channel. When set to the left channel, the data will be output following WS's falling edge and when set to output on the right channel, data will be output following WS's rising edge.

For a stereo application, the SD pins of the left and right ICS-43434 microphones should be tied together as shown in Figure 8. The format of a stereo I²S data stream is shown in Figure 9. Figure 10 and Figure 11 show the formats of a mono microphone data stream for left and right microphones, respectively.

Data Output Mode

The output data pin (SD) is tristated when it is not actively driving I²S output data. SD immediately tristates after the LSB is output so that another microphone can drive the common data line.

The SD trace should have a pull-down resistor to discharge the line during the time that all microphones on the bus have tristated their outputs. A 100 k Ω resistor is sufficient for this, as shown in Figure 8. If the SD line needs to be discharged faster than a 100 k Ω resistor can, a smaller resistor, such as 10 k Ω , can be used.

Data Word Length

The output data word length is 24 bits per channel.

Data Word Format

The default data format is I²S (twos complement), MSB-first. In this format, the MSB of each word is delayed by one SCK cycle from the start of each half-frame.

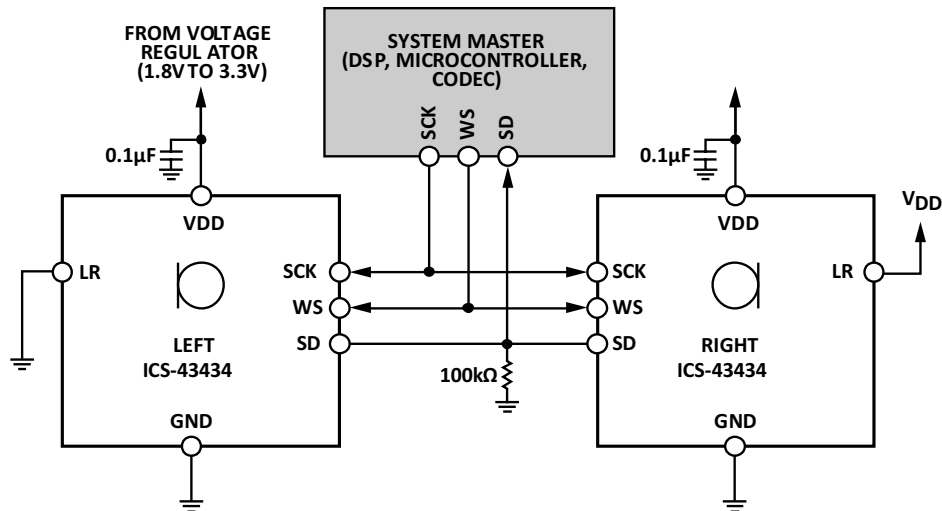


Figure 8. System Block Diagram

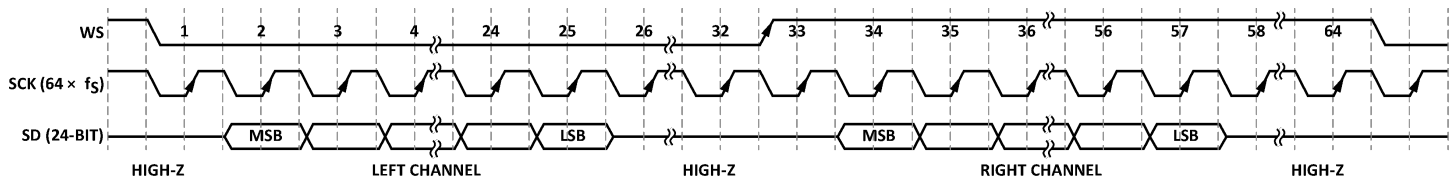


Figure 9. Stereo Output I²S Format

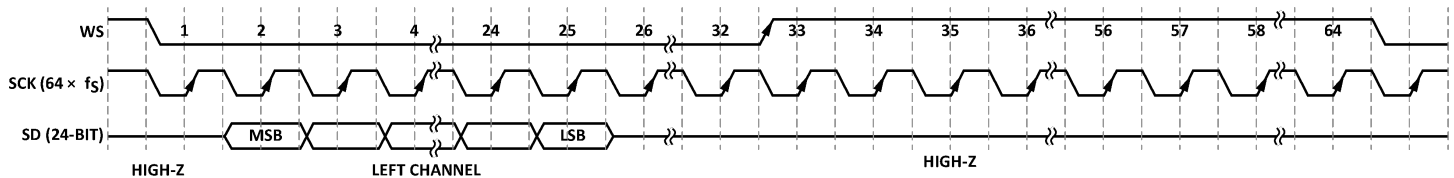


Figure 10. Mono Output I²S Format Left Channel (LR = 0)

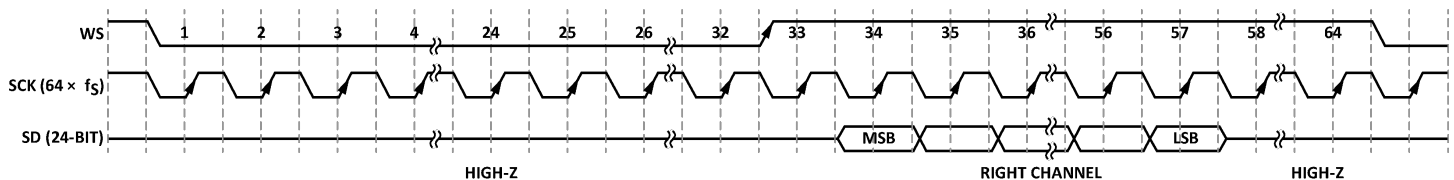


Figure 11. Mono Output I²S Format Right Channel (LR = 1)

Data Output Format

The output data word length is 24 bits/channel. The data word format is 2's complement, MSB-first.

The output data pin (SD) is tri-stated when it is not actively driving output data. SD will immediately tri-state after the LSB is output so that another microphone can drive the common data line.

DIGITAL MICROPHONE SENSITIVITY

The sensitivity of a digital output microphone is specified in units of dB FS (decibels relative to a full-scale digital output). A 0 dB FS sine wave is defined as a signal whose peak just touches the full-scale code of the digital word (see Figure 5). This measurement convention means that signals with a different crest factor may have an RMS level higher than 0 dB FS. For example, a full-scale square wave has an RMS level of 3 dB FS.

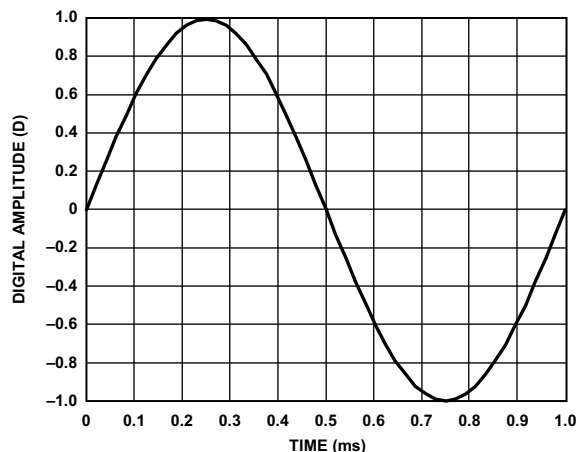


Figure 11. 1 kHz, 0 dB FS Sine Wave

The definition of a 0 dB FS signal must be understood when measuring the sensitivity of the ICS-43434. An acoustic input signal of a 1 kHz sine wave at 94 dB SPL applied to the ICS-43434 results in an output signal with a -26 dB FS level. This means that the output digital word peaks at -26 dB below the digital full-scale level. A common misunderstanding is that the output has an RMS level of -29 dB FS; however, this is not the case because of the definition of a 0 dB FS sine wave.

There is no commonly accepted unit of measurement to express the instantaneous level of a digital signal output from the microphone, as opposed to the RMS level of the signal. Some measurement systems express the instantaneous level of an individual sample in units of D, where 1.0 D is digital full scale (see Figure 11). In this case, a -26 dB FS sine wave has peaks at 0.05 D.

For more information about digital microphone sensitivity, see the AN-1112 Application Note, *Microphone Specifications Explained*.

DIGITAL FILTER CHARACTERISTICS

The ICS-43434 has an internal digital bandpass filter. A high-pass filter eliminates unwanted low frequency signals. A low-pass decimation filter scales the pass band with the sampling frequency and performs required out-of-band noise reduction.

High-Pass Filter

The ICS-43434 incorporates a high-pass filter to remove DC and low frequency components. This high pass filter has a -3 dB corner frequency of 24 Hz and does not scale with the sampling rate.

Low-Pass Decimation Filter

The analog-to-digital converter in the ICS-43434 is a single-bit, high order, sigma-delta ($\Sigma\Delta$) running at a high oversampling ratio. The noise shaping of the converter pushes the majority of the noise well above the audio band and gives the microphone a wide dynamic range. However, it does require a good quality low-pass decimation filter to eliminate the high frequency noise.

The pass band of the filter extends to $0.417 \times f_s$ and, in that band, has only ± 0.3 dB of ripple. A 48 kHz sampling rate results in a pass band of 20 kHz and a half amplitude corner at 24 kHz; the stop-band attenuation of the filter is 58 dB. Note that these filter specifications scale with sampling frequency.

APPLICATIONS INFORMATION

LOW-POWER MODE

Low Power Mode (LPM) enables the ICS-43434 to be used in an AlwaysOn listening mode for keyword spotting and ambient sound analysis. The ICS-43434 will enter LPM when the sampling frequency is between 6.25 and 18.75 kHz. In this mode, the microphone consumes only 230 μA while retaining high electro-acoustic performance.

When one microphone is in LPM for AlwaysOn listening, a second microphone sharing the same data line may be powered down. In this case, where one microphone is powered up and another is powered down by disabling the VDD supply or in sleep mode by reducing the frequency of a separate clock source, the disabled microphone does not present a load to the signal on the LPM microphone's DATA pin.

SLEEP MODE

The microphone enters sleep mode when the sampling frequency falls below 3.125 kHz. In this mode, the microphone data output is in a high impedance state. The current consumption in sleep mode is 12 μA .

The ICS-43434 enters sleep mode within 1 ms of the sampling frequency falling below 3.125 kHz. The microphone wakes up from sleep mode and begins to output data 32,768 SCK cycles after the clock becomes active. For a 3.072 MHz clock ($f_s = 48 \text{ kHz}$), the microphone starts to output data in 10.7 ms. For a 2.4 MHz clock ($f_s = 37.5 \text{ kHz}$), the microphone starts to output data in 13.7 ms. The wake-up time (Table 5) indicates the time from when the clock is enabled to when the ICS-43434 outputs data within 1 dB of its settled sensitivity.

SD OUTPUT DRIVE STRENGTH

The SD data output pin must drive a load that includes the PCB trace and the tri-stated inputs of the other ICS-43434 SD pins connected to that same trace. The tri-stated load capacitance of the ICS-43434 SD pin is about 6 pF. The ICS-43434 has been designed to drive a load of 85 pF.

POWER SUPPLY DECOUPLING

For best performance and to avoid potential parasitic artifacts, placing a 0.1 μF ceramic type X7R or better capacitor between Pin 5 (VDD) and ground is strongly recommended. The capacitor should be placed as close to Pin 3 as possible.

The connections to each side of the capacitor should be as short as possible, and the trace should stay on a single layer with no vias. For maximum effectiveness, locate the capacitor equidistant from the power and ground pins or, when equidistant placement is not possible, slightly closer to the power pin. Thermal connections to the ground planes should be made on the far side of the capacitor, as shown in Figure 12.

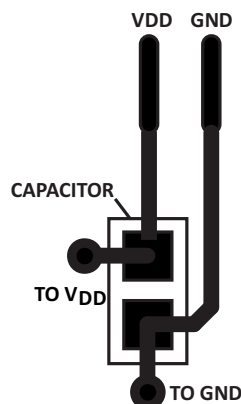


Figure 12. Recommended Power Supply Bypass Capacitor Layout

SUPPORTING DOCUMENTS

For additional information, see the following documents.

EVALUATION BOARD USER GUIDE

AN-000088, *Bottom-Port I²S Output MEMS Microphone Evaluation Board*

APPLICATION NOTES

AN-100, *MEMS Microphone Handling and Assembly Guide*

AN-1003, *Recommendations for Mounting and Connecting the InvenSense Bottom-Ported MEMS Microphones*

AN-1112, *Microphone Specifications Explained*

AN-1124, *Recommendations for Sealing InvenSense Bottom-Port MEMS Microphones from Dust and Liquid Ingress*

AN-1140, *Microphone Array Beamforming*

PCB DESIGN AND LAND PATTERN LAYOUT

The recommended PCB land pattern for the ICS-43434 should be laid out to a 1:1 ratio to the solder pads on the microphone package, as shown in Figure 13. Take care to avoid applying solder paste to the sound hole in the PCB. A suggested solder paste stencil pattern layout is shown in Figure 14. The diameter of the sound hole in the PCB should be larger than the diameter of the sound port of the microphone. A minimum diameter of 0.5 mm is recommended.

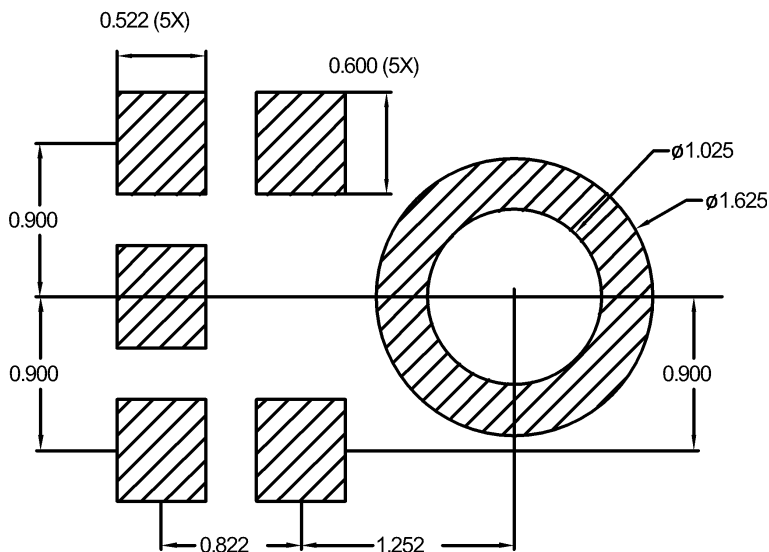


Figure 13. PCB Land Pattern Layout

Dimensions shown in millimeters

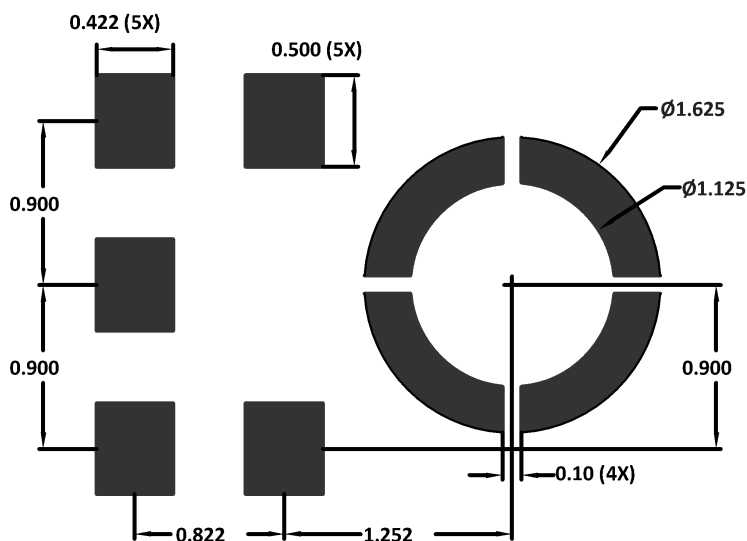


Figure 14. Suggested Solder Paste Stencil Pattern Layout

Dimensions shown in millimeters

PCB MATERIAL AND THICKNESS

The performance of the ICS-43434 is not affected by PCB thickness. The ICS-43434 can be mounted on either a rigid or flexible PCB. A flexible PCB with the microphone can be attached directly to the device housing with an adhesive layer. This mounting method offers a reliable seal around the sound port while providing the shortest acoustic path for good sound quality.

HANDLING INSTRUCTIONS

PICK AND PLACE EQUIPMENT

The MEMS microphone can be handled using standard pick-and-place and chip shooting equipment. Take care to avoid damage to the MEMS microphone structure as follows:

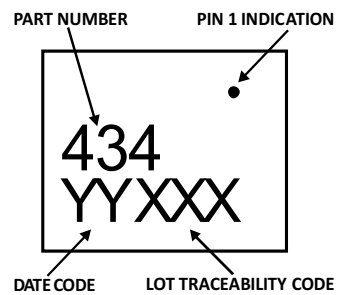
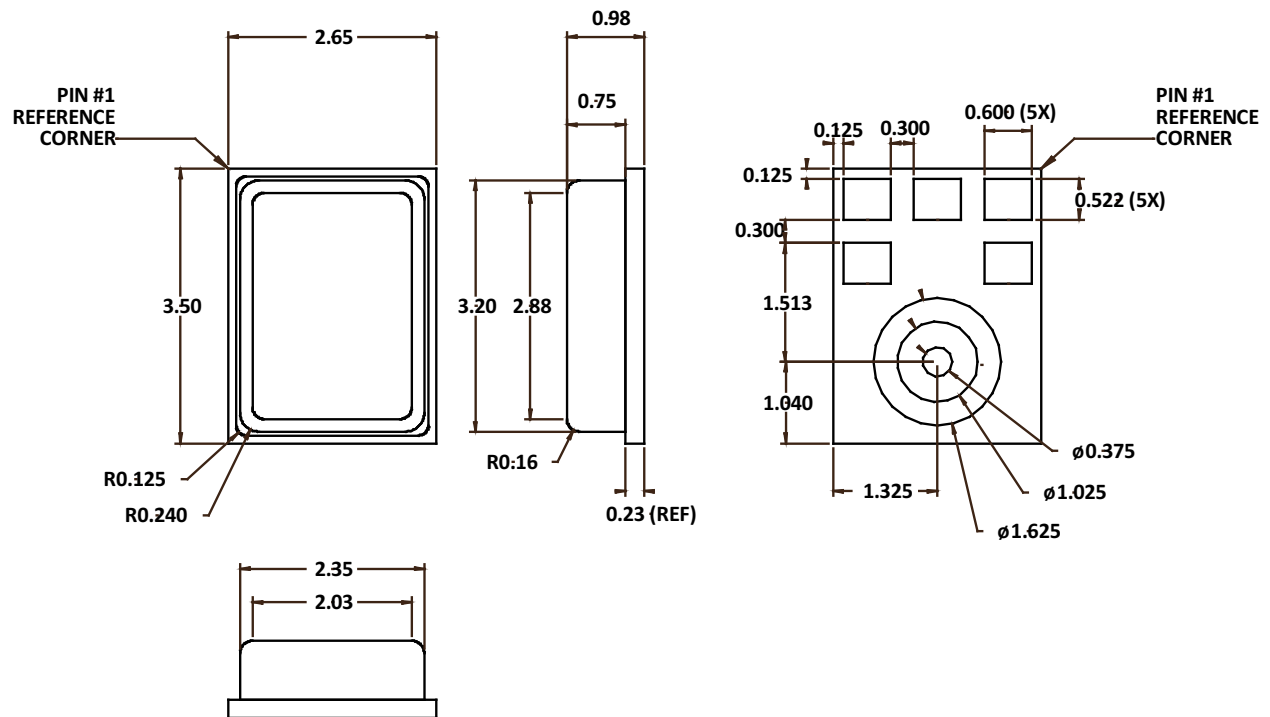
- Use a standard pickup tool to handle the microphone. Because the microphone hole is on the bottom of the package, the pickup tool can make contact with any part of the lid surface.
- Do not pick up the microphone with a vacuum tool that makes contact with the bottom side of the microphone. Do not pull air out of or blow air into the microphone port.
- Do not use excessive force to place the microphone on the PCB.

REFLOW SOLDER

For best results, the soldering profile must be in accordance with the recommendations of the manufacturer of the solder paste used to attach the MEMS microphone to the PCB. It is recommended that the solder reflow profile not exceed the limit conditions specified in Figure 2 and Table 7.

BOARD WASH

When washing the PCB, ensure that water does not make contact with the microphone port. Do not use blow-off procedures or ultrasonic cleaning.



PART	TEMP RANGE	PACKAGE	QUANTITY	PACKAGING
ICS-43434	-40°C to +85°C	6-Terminal LGA_CAV	10,000	13" Tape and Reel
EV_ICs-43434-FX		Flex Evaluation Board		

REVISION HISTORY

REVISION DATE	REVISION	DESCRIPTION
4/26/2016	1.0	Initial version
6/3/2016	1.1	Updated Ordering Quantity
8/29/2016	1.2	Updated HPM SNR, EIN and Dynamic Range value

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